

Introduction to Excel

BASIC FUNCTIONALITY



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Excel Basics

- Microsoft Excel is a spreadsheet application developed by Microsoft for Microsoft Windows and Mac OS. It features calculation, graphing tools, pivot tables, and a macro programming language called Visual Basic for Applications.

- Microsoft Excel has the basic features of all spreadsheets, using a grid of cells arranged in numbered rows and letter-named columns to organize data manipulations like arithmetic operations. It has a battery of supplied functions to answer statistical, engineering and financial needs. In addition, it can display data as line graphs, histograms and charts, and with a very limited three-dimensional graphical display.

4

Excel Basics

- Ribbon- Topleft of the ribbon has the option “File” on the menu



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- Within File menu, you have

‘Save’ – to save the current open file

‘Save as’ – to save the current open file with a different name or extension

‘Open’ - to open a new file

‘Close’ – to close the current file

‘Recent’ – to List of recently opened files

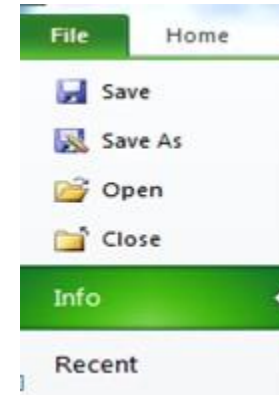
‘New’ – to Open a new file

‘Print’ – to Print the currently open file

‘Save & Send’ – if you want to save and email the currently open file

‘Help’ – any help required on Excel

‘Exit’ – To exit the program



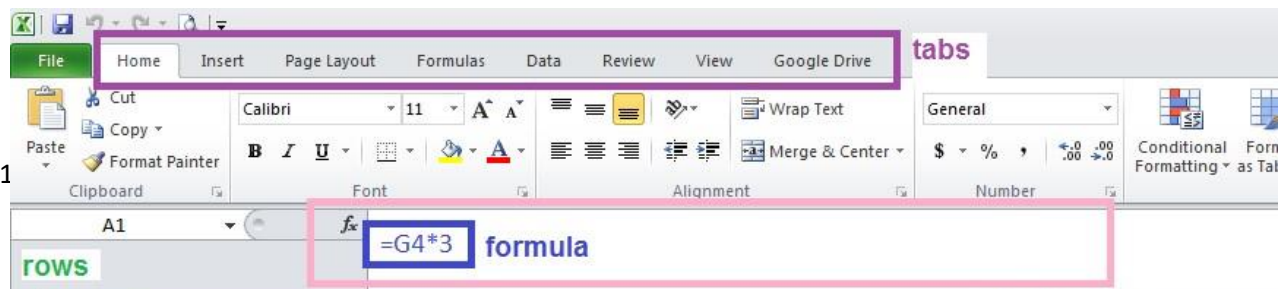
Excel Spreadsheet – First look

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Version 1



Excel Rows and Columns

Spreadsheets are displayed in a grid layout. The letters across the top header are Column headings. To highlight an entire Column, click on any of the letters.

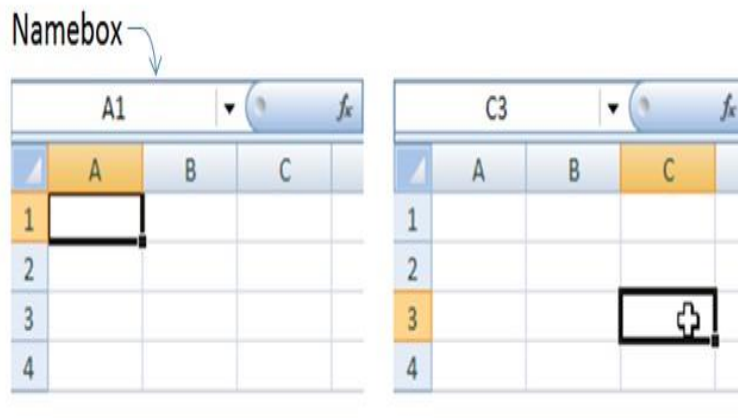
Similarly, Rows are defined like 1,2,3.

Spreadsheets are all about individual cells. A Cell is an intersection of a column and a row and is named by combining a letter with a number.

So if you combine the B column with Row 5, you get Cell B5. Combine Column D with Row 5 and you get Cell D5. The cell name is shown in the “Namebox”

Excel Rows and Columns

To see this for yourself, click inside any of the cells on your spreadsheet. In the Images below, we have clicked inside cell A1 and cell C3.



The first picture is Column A, Row 1 (A1), and the second picture is Column C Row 3 (C3). Notice that the cells we clicked on have a black border around them. This tells you the cell is active.

Excel Rows and Columns

- The cell that is active will have its Column letter and Row number displayed in the top left box called “Namebox”, just above the letters A and B in the pictures. When you click into a cell, you can then type text and numbers.
- To move around the spreadsheet, and make other cells active, you can either just click inside a Cell, or press the arrow keys on your keyboard. Click inside a Cell and notice the Cell reference appear in the Namebox.
- Press your arrow keys and notice how the active cells moves. Before going any further, make sure you understand how the spreadsheet grid works. If you are asked to locate Cell H2, you should be able to do so.

- To enter a value in a cell, just click on it twice, this will activate the cell.

Formatting the font

Excel has a much wider range of formatting options now than the previous versions, and it's relatively easy to turn a dull spreadsheet into something that really shines.

- Choosing a Font in Excel - You can pick a different font for the data you enter into cells, as well as choosing the size you want.
- The color of the font, and the cell background, can be changed, too.
- In Excel, Themes have been introduced, so that you can format your spreadsheets more easily.

First, we'll see how to change the font type.

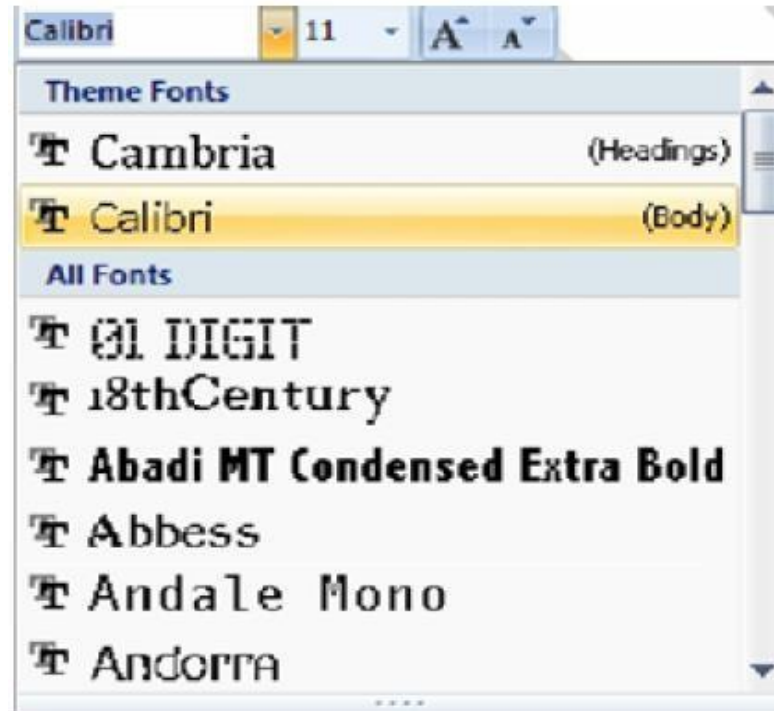
Highlight cell A1 on your spreadsheet by simply clicking into it. Locate the Font panel on the Excel Ribbon at the top of the page



The font in the panel above is set to Calibri by default in the newer versions of MS Excel.

The good thing about Excel is that when you move your mouse over one of the fonts on the list, the text in your selected cell (A1) will change automatically. This is just a preview, though. When you have decided on the font you want, click it with the left mouse button.

To see more fonts, click the black dropdown:



In a similar way, font size can be changed by clicking on the number next to fonts dropdown.

Working with Data

Working with data is one of the prime objectives of using an Excel workbook. There are some basic features available in Excel to ease the data workings.

- Selecting a cell
- Freezing the panes
- Sorting the data
- Plotting the data

Selecting a cell

	A	B	C
1			
2			
3			
4			

Deselected cells look like this

In order to select one cell, the simplest way is to click on it. When you click on it, the cell borders get highlighted like this

	A	B	C
1			
2			
3			

Here, cell A1 has been selected by clicking on it.

Now, let's say we want to select the cells from A1 to C3. Any ideas how do you do so?

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Selecting a cell

It's simple!

	A	B	C	D
1				
2				
3				
4				

Just, click on A1 cell.

Now, move the selection using “<Shift> + Arrow keys” to move the selection towards right and down.

	A	B	C	D
1				
2				
3				
4				
5				

*If you move the selection only with Arrow keys, the A1 cell will be deselected.
While if you use*

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Arrow keys along with Shift key, the selection will expand. Also, you can use “<Shift> +<Ctrl> +Arrow keys” if the number of rows and columns is large. This shortcut will select the entire range of columns and rows. Try it out!

Freezing the panes

While doing a back-testing for a strategy, normally the span of data considered is for years. So, it becomes difficult to view the entire dataset in one view. Even if we try to view by scrolling the data, the headers are lost. What’s the solution then? **Freezing the pane!**

Let’s take an example of closing price data. When we scroll down, the top for “Yahoo!” stock row is not visible.

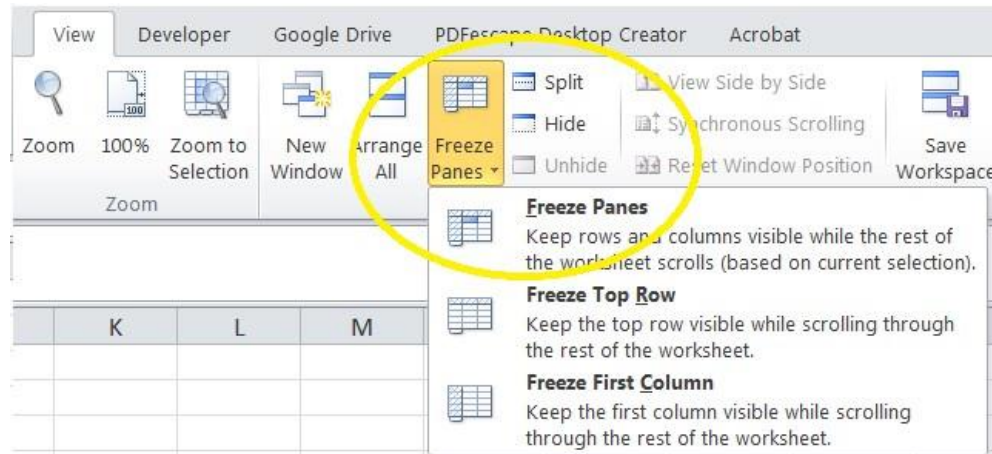
	A	B
1	Date	Adj Close
2	6/4/2015	26813.41992
3	6/3/2015	26837.19922
4	6/2/2015	27188.38086
5	6/1/2015	27848.99023
6	5/29/2015	27828.43945
7	5/28/2015	27506.71094
8	5/27/2015	27564.66016
9	5/26/2015	27531.41016
10	5/25/2015	27643.88086
11	5/22/2015	27957.5
12	5/21/2015	27809.34961
13	5/20/2015	27837.21094
14	5/19/2015	27645.5293
15	5/18/2015	27687.30078
16	5/15/2015	27324
17	5/14/2015	27206.06055
18	5/13/2015	27251.09961
19	5/12/2015	26877.48047
20	5/11/2015	27507.30078
21	5/8/2015	27105.39062
22	5/7/2015	26599.10938

12	5/21/2015	27809.34961
13	5/20/2015	27837.21094
14	5/19/2015	27645.5293
15	5/18/2015	27687.30078
16	5/15/2015	27324
17	5/14/2015	27206.06055
18	5/13/2015	27251.09961
19	5/12/2015	26877.48047
20	5/11/2015	27507.30078
21	5/8/2015	27105.39062
22	5/7/2015	26599.10938
23	5/6/2015	26717.36914
24	5/5/2015	27440.14062
25	5/4/2015	27490.58984
26	5/1/2015	27011.31055

Freezing the panes

On the View tab, in the Window group, click the arrow below Freeze Panes.

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There are three options available:

Freeze Panes

Freeze Top Row

Freeze First Column

	A	B
1	Date	Adj Close
113	12/24/2014	27208.60938
114	12/23/2014	27506.46094
115	12/22/2014	27701.78906
116	12/19/2014	27371.83984
117	12/18/2014	27126.57031
118	12/17/2014	26710.13086
119	12/16/2014	26781.43945
120	12/15/2014	27319.56055
121	12/12/2014	27350.67969
122	12/11/2014	27602.00977
123	12/10/2014	27831.09961
124	12/9/2014	27797.00977
125	12/8/2014	28119.40039
126	12/5/2014	28458.09961
127	12/4/2014	28562.82031
128	12/3/2014	28442.71094
129	12/2/2014	28444.00977

Freeze Top Row

Let's see what happens when we select the option 'Freeze Top Row'. Steps are:

1. Select the Dataset for which we want to apply this (by selecting the cells)
2. Click on the "Freeze Top Row" from the ribbon.
3. Now, even if we scroll to the end of the data set the top row would still be visible as shown in the image right.

Do try how "Freeze First Column" on your own with the data provided

	A	B
1	Date	Adj Close
113	12/24/2014	27208.60938
114	12/23/2014	27506.46094
115	12/22/2014	27701.78906
116	12/19/2014	27371.83984
117	12/18/2014	27126.57031
118	12/17/2014	26710.13086
119	12/16/2014	26781.43945
120	12/15/2014	27319.56055
121	12/12/2014	27350.67969
122	12/11/2014	27602.00977
123	12/10/2014	27831.09961
124	12/9/2014	27797.00977
125	12/8/2014	28119.40039
126	12/5/2014	28458.09961
127	12/4/2014	28562.82031
128	12/3/2014	28442.71094
129	12/2/2014	28444.00977

on

Freeze Panes

To freeze more than one row or column, we use “Freeze Panes”. It allows us to freeze as many rows and columns as we want.

Here, we select the rows and columns up to which we want to freeze view, i.e.

- We select Row 4 and Column D, the excel pane will freeze from Row 1 to Row 4 and Column A to Column D,
- Effectively, all the data will always be visible from A1 to D4 irrespective of the extent of scroll done.

Try this on your own, and see the result!

Sorting the data

While back-testing a strategy, we need to have historical data for the stock. This data is generally has the pattern of latest date at the top. Usually a strategy written in Excel requires the data in reverse chronological order, i.e. oldest at the top.

Downloaded format

	A	B	
1	Date	Adj Close	
2	14-12-2017	10159.55	
3	13-12-2017	10192.95	
4	12-12-2017	10240.15	
5	11-12-2017	10322.25	
6	08-12-2017	10265.65	
7	07-12-2017	10166.7	
8	06-12-2017	10044.1	

Required format

	A	B	
1	Date	Adj Close	
2	06-12-2017	10044.1	
3	07-12-2017	10166.7	
4	08-12-2017	10265.65	
5	11-12-2017	10322.25	
6	12-12-2017	10240.15	
7	13-12-2017	10192.95	
8	14-12-2017	10159.55	

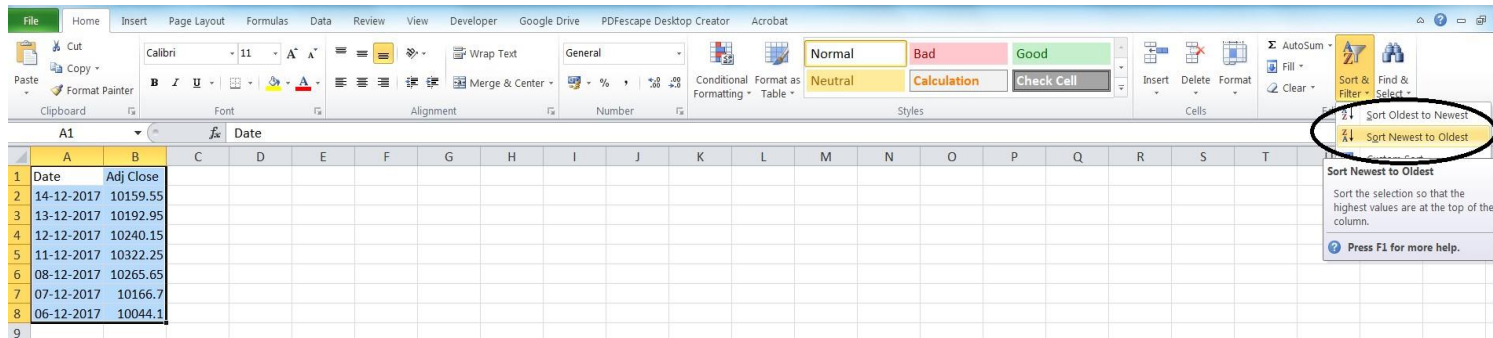
This can be done in multiple ways in Excel, the easiest one being the one explained on next slide.

Sorting the data

All you need to do is,

Select the cells using “<Shift>+<Ctrl>+Arrow Keys” as taught in earlier slides

Click the Home tab on the ribbon, and then click on “Sort & Filter”



And the result will be

	A	B	C	D	E	F
1	Date	Adj Close				
2	06-12-2017	10044.1				
3	07-12-2017	10166.7				
4	08-12-2017	10265.65				
5	11-12-2017	10322.25				
6	12-12-2017	10240.15				
7	13-12-2017	10192.95				
8	14-12-2017	10159.55				
9						

Sorting the data

Have you noticed that, the date format was dd-mm-yyyy i.e. 14-12-2017 or 14 Dec 2017.

Did you wonder, if the same Sorting process will work if the date format was 14th December 2017. Why don't you try this?

Following is the data

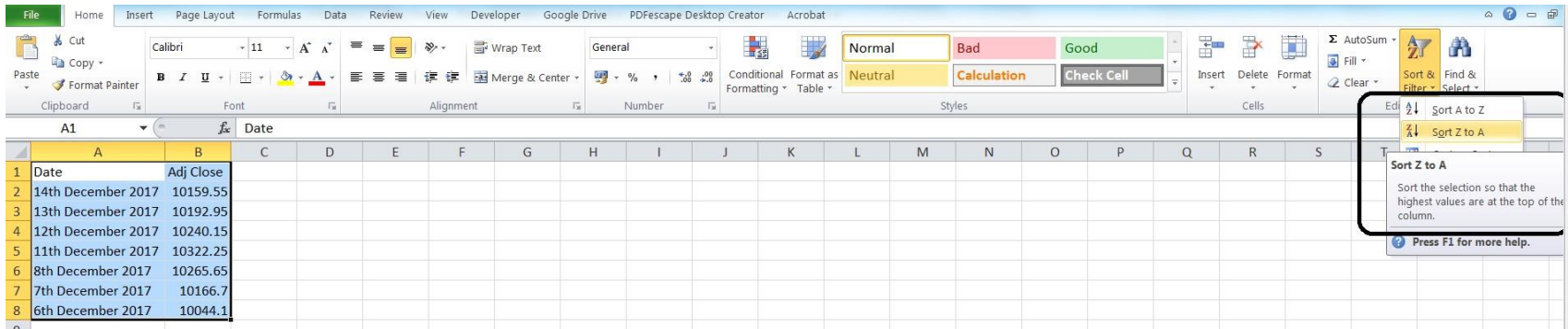
	A	B	C
1	Date	Adj Close	
2	14th December 2017	10159.55	
3	13th December 2017	10192.95	
4	12th December 2017	10240.15	
5	11th December 2017	10322.25	
6	8th December 2017	10265.65	
7	7th December 2017	10166.7	
8	6th December 2017	10044.1	
9			

Sorting the data

If we apply the same sorting process, can you see the difference in options Excel is providing you with?

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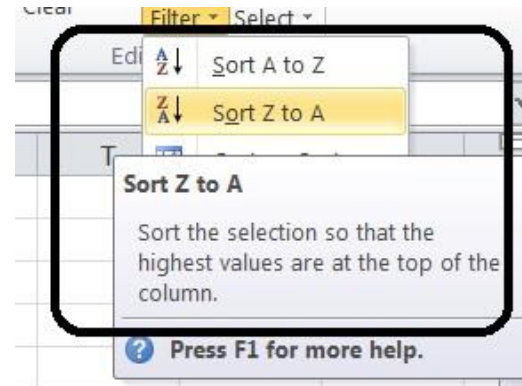




Earlier, the options were



and now



Sorting the data

Why this difference?

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Answer to this question lies in the difference of the format of data. In first example, the data was in Date format, hence Excel gave you the options for “Oldest to Newest” and “Newest to Oldest”

While in second, the format was normal text. Hence, the options, “A to Z” and “Z to A”.

Now, what to do in second scenario, where format is Text. Nothing to worry about, Excel is such a friendly tool, just change the format of data! That’s again simple, just explore a bit and you will know!

Plotting Data – Creating Graphs

A **chart**, or **graph**, is a visual representation of a set of data. And is an important tool for representing data in an analysis.

There are multiple kind of charts available in Excel for example – Line Graph, Scatter chart, Column graph, Pie Chart, etc. You must be wondering where can you find them? Here is the answer!

If you clickon the “Insert” Tab in the Ribbon,



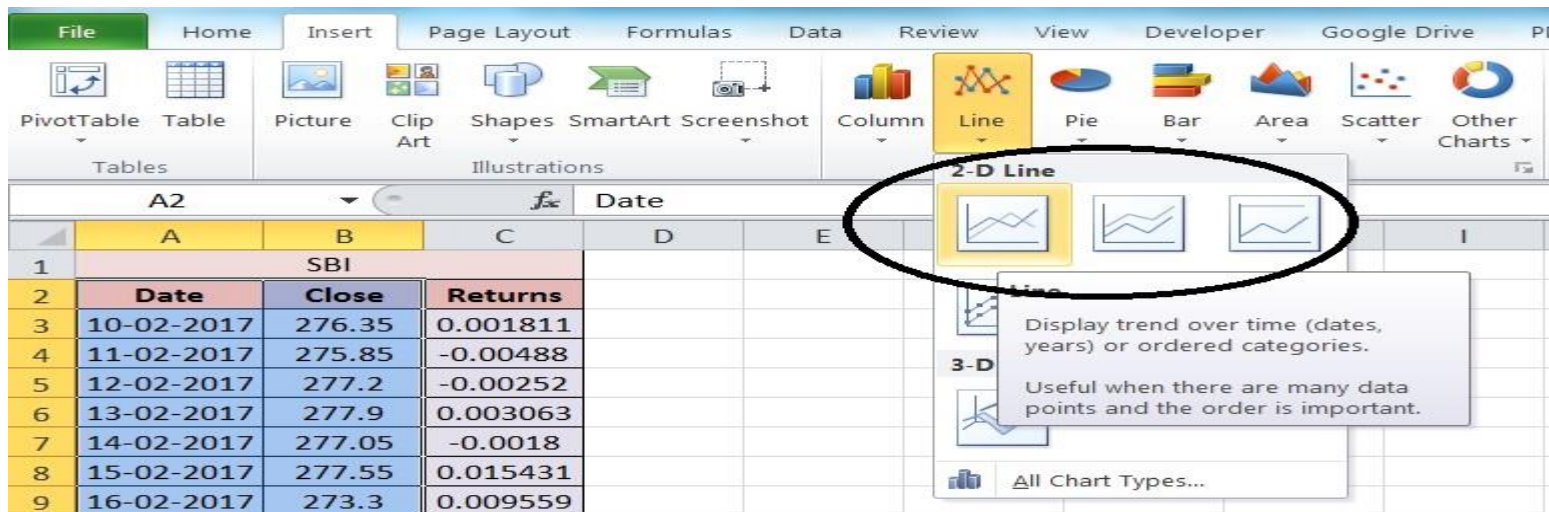
You can see the charts mentioned, “Column”, “Line”, “Pie”, “Bar”, “Area”, “Scatter”, and “Others”.

Let’s learn how to plot the Line chart!

Plotting Data – Creating Graphs

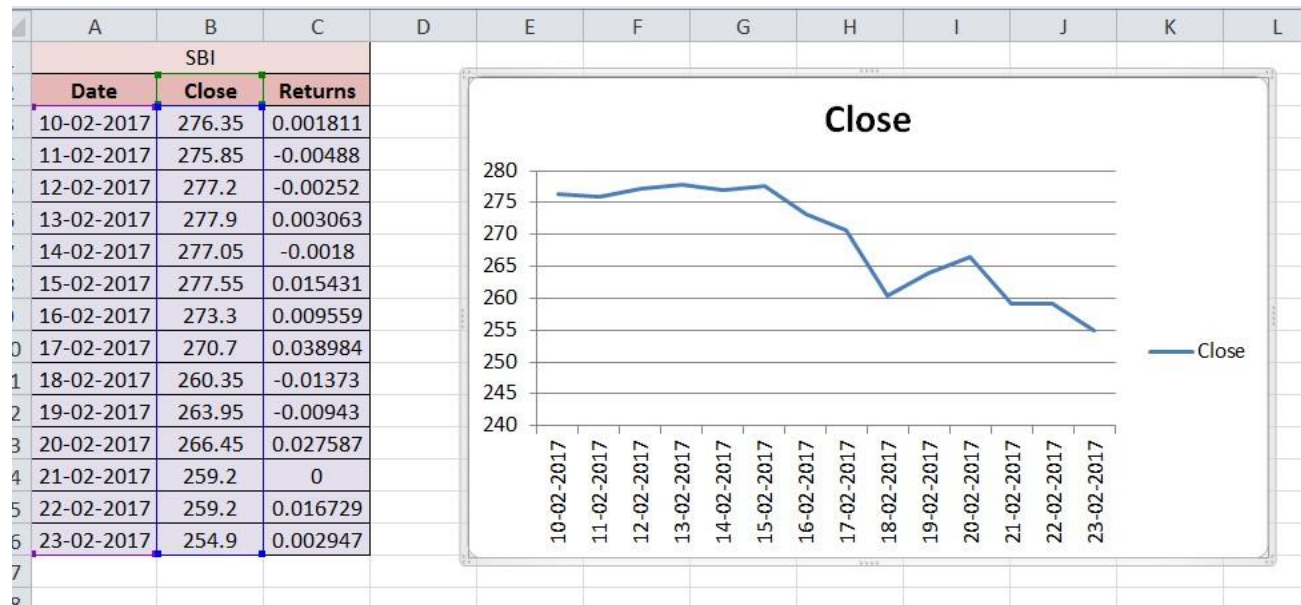
To plot a Line chart, follow the steps given below:

- Select the data source with the range of data you want to chart. Data selection is done exactly the same way Cell selection is done.
- In the Charts group on the Insert tab, click the “Line” chart type, and then click a chart subtype in the Chart gallery.



Plotting – Creating graphs

Choose the line type and select 2D line as this is a suitable one for time series data.



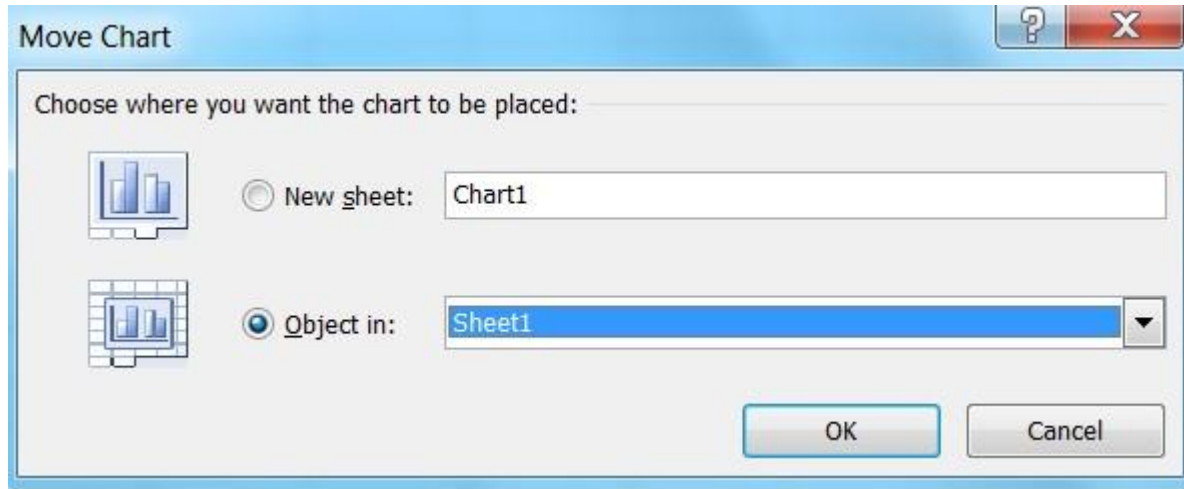
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In the Location group on the Chart Tools Design tab, click the Move Chart button to place the chart in a chart sheet or embed it into a worksheet.

Plotting – Creating graphs



When you click on Move chart, you will get a pop up



Where you can select the new position of the chart either in same sheet or other sheets.

Plotting – Creating graphs

Another important chart for a trader or Quant Analyst is Histogram.

What is a Histogram?

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A histogram is an accurate representation of the distribution of numerical data. It is an estimate of the probability distribution of a continuous variable (quantitative variable). It is a kind of bar graph.

To construct a histogram, the first step is to "bin" the range of values—that is, divide the entire range of values into a series of intervals—and then count how many values fall into each interval.

The bins are usually specified as consecutive, non-overlapping intervals of a variable. The bins (intervals) must be adjacent, and are often (but are not required to be) of equal size.

Definition Source: Wikipedia

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Analysis ToolPak Add-in

While doing the back testing we do need Histogram Analysis in order to identify.

Now, how to create a Histogram in an Excel?

Excel provides a Histogram analysis tool which can be used to generate a Histogram Analysis. To do so, you first need to make sure that the Analysis ToolPak Add-in is installed.

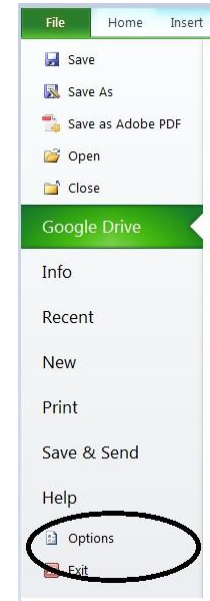
Steps to install Analysis ToolPak Add-in in Excel 2010:

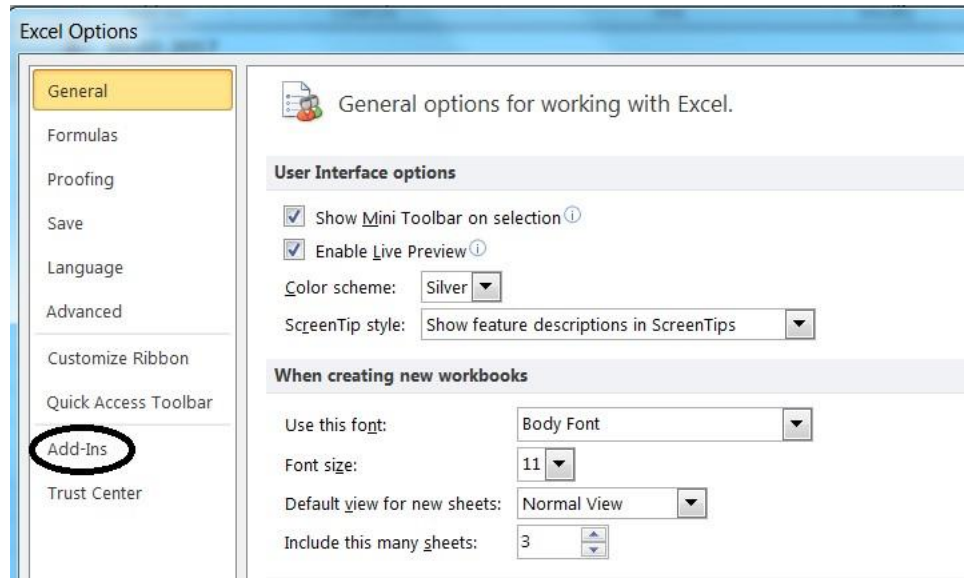
On **File** menu, click on **Options**

Analysis ToolPak Add-in - Installation

When you click on Options, a pop-up comes up, there you
Ins

click on “**Add-**





Analysis ToolPak Add-in - Installation

Then, select “Analysis ToolPak” from the list, and click on “Go”, once you have selected Excel Add-ins.

Inactive Application Add-ins

Analysis ToolPak	C:\...ft Office\Office14\Library\Analysis\ANALYS32.XLL	Excel Add-in
Analysis ToolPak - VBA	C:\...ft Office\Office14\Library\Analysis\ATPVBAFMT.VBA	Excel Add-in
Custom XML Data	C:\...ram Files\Microsoft Office\Office14\OFFRHD.DLL	Document Inspector
Euro Currency Tools	C:\...icrosoft Office\Office14\Library\EUROTOOL.XLAM	Excel Add-in
Financial Symbol (XML)	C:\...on Files\Microsoft Shared\Smart Tag\MOFL.DLL	Action
Headers and Footers	C:\...ram Files\Microsoft Office\Office14\OFFRHD.DLL	Document Inspector
Hidden Rows and Columns	C:\...ram Files\Microsoft Office\Office14\OFFRHD.DLL	Document Inspector
Hidden Worksheets	C:\...ram Files\Microsoft Office\Office14\OFFRHD.DLL	Document Inspector
Invisible Content	C:\...ram Files\Microsoft Office\Office14\OFFRHD.DLL	Document Inspector
Microsoft Actions Pane 3		XML Expansion Pack
Solver Add-in	C:\...ft Office\Office14\Library\SOLVER\SOLVER.XLAM	Excel Add-in

Document Related Add-ins
No Document Related Add-ins

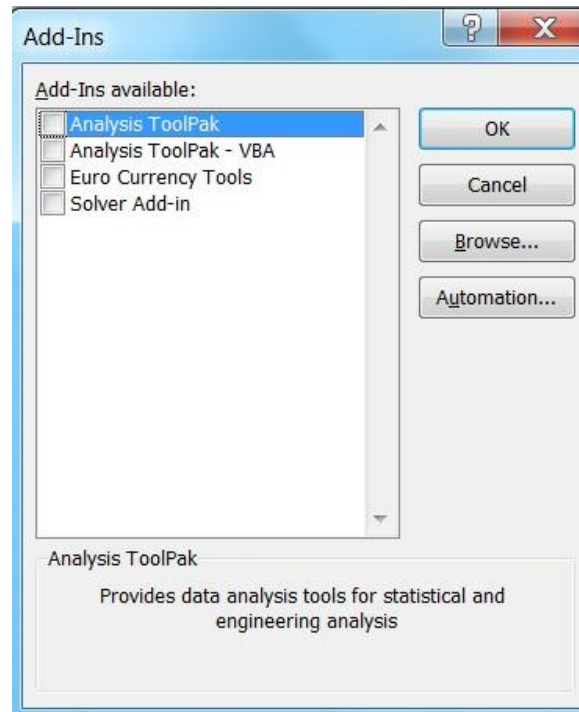
Disabled Application Add-ins

Add-in: Acrobat PDFMaker Office COM Addin
 Publisher: Adobe Systems, Incorporated
 Compatibility: No compatibility information available
 Location: C:\Program Files (x86)\Adobe\Acrobat DC\PDFMaker\Office\x64\PDFMOfficeAddin.dll
 Description: Acrobat PDFMaker Office COM Addin

Manage: Excel Add-ins ▼ Go...

Analysis ToolPak Add-in - Installation

A pop-up will come for Add-Ins where you select “Analysis ToolPak” and Click on “OK”



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This will enable the “Data Analysis” feature in Excel under “Data” tab on the ribbon.

Analysis ToolPak Add-in

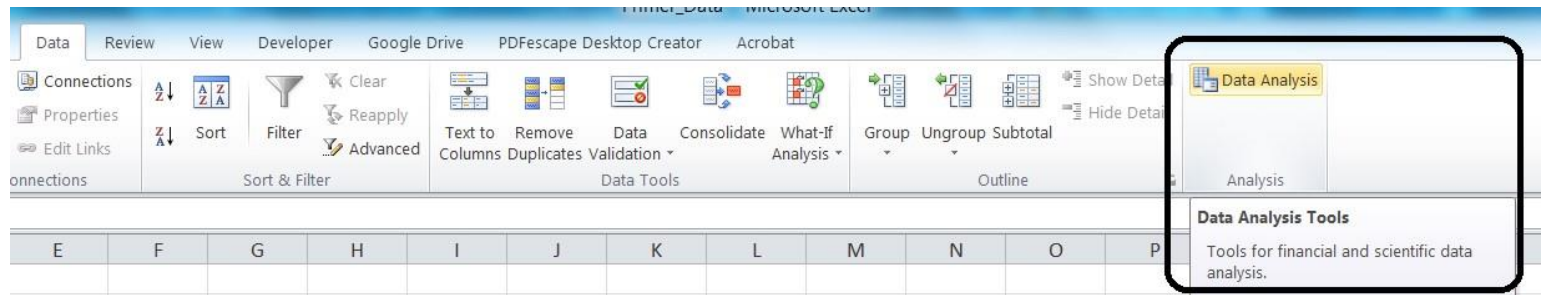
How to use Data Analysis feature?

Data Analysis feature creates the Histogram for you in just two clicks! Want to know how? Just follow the steps:

Once the Data Analysis feature is activated/installed, select the data for which you want the Histogram Analysis;

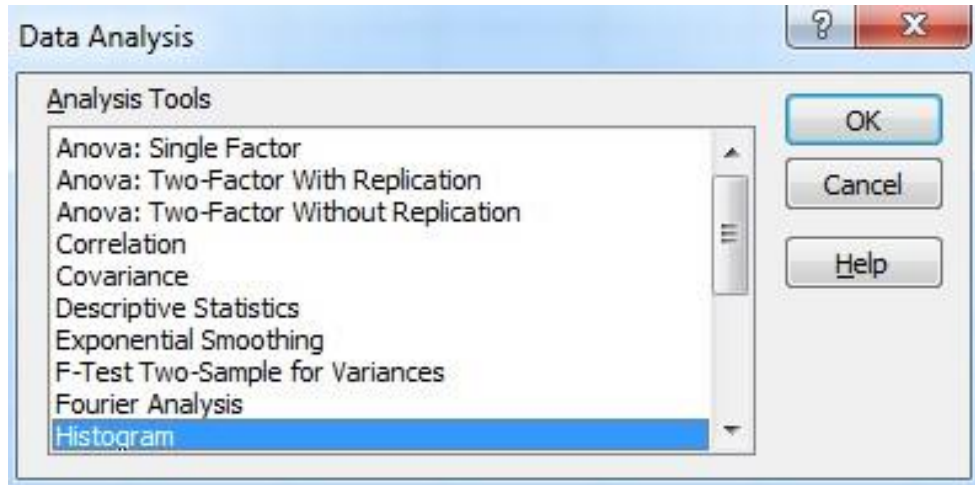
Click on “Data Analysis” icon in the Excel ribbon under “Data” tab.

Date	Close
10-02-2017	276.35
11-02-2017	275.85
12-02-2017	277.2
13-02-2017	277.9
14-02-2017	277.05
15-02-2017	277.55
16-02-2017	273.3
17-02-2017	270.7
18-02-2017	260.35
19-02-2017	263.95
20-02-2017	266.45
21-02-2017	259.2
22-02-2017	259.2
23-02-2017	254.9



Creating Histogram

On clicking “Data Analysis” you will get a pop-up as shown below



Select Histogram and click “OK”.

This will ask you to provide the Input range and the Bin range for the Histogram along with the output options.

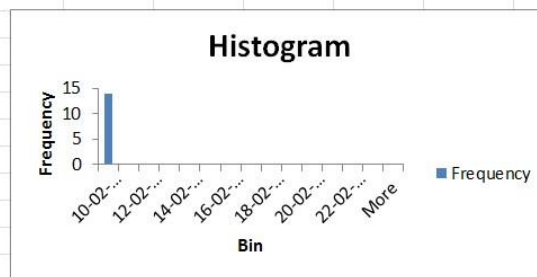
Creating Histogram

Give Input range as “\$B\$3:\$B\$16” while Bin range as “\$A\$3:\$A\$16”

Output options as New Workbook and select

“Chart Output” ; and click on “OK”
This will generate the Histogram in a new workbook as shown below

1	Bin	Frequency
2	10-02-2017	14
3	11-02-2017	0
4	12-02-2017	0
5	13-02-2017	0
6	14-02-2017	0
7	15-02-2017	0
8	16-02-2017	0
9	17-02-2017	0
10	18-02-2017	0
11	19-02-2017	0
12	20-02-2017	0
13	21-02-2017	0
14	22-02-2017	0
15	23-02-2017	0
16	More	0



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Here, the data not a distributed one, hence the frequency is for only one instance and rest as 0.

Though creating a histogram is easy in Excel, but the best the easiest way is to create with Python. You will learn the same during your EPAT lectures!

Plotting – Summary

Now, let's go through the utility of different types of Charts in Algo domain.

Though there are multiple types of charts, but here we will compare the ones used most frequently in trading.

Chart Name	Utility
Line Charts	Line charts can show continuous data over time on an evenly scaled Axis. Therefore, they are ideal for showing trends in data at equal intervals, such as months, quarters or years.

XY (Scatter) charts	XY (Scatter) charts are typically used for showing and comparing numeric values, like scientific, statistical, and engineering data. Scatter plots show how much one variable is affected by another. The relationship between two variables is called their correlation
Histogram	When you want to show the frequency data.

Referencing a Cell

In Excel, we can refer a cell, instead of always entering the proper data value.

Referencing is a very important tool, used to 'refer' to a value stored elsewhere.

For example, if $C2 = B2/B1$, then C2 is quotient of B2 divided by B1. This is **Relative Reference**.

Another example, if $C2 = B2/B\$1$ and we copy C2 formula in all the rows of column 'C', then column 'C' consists of quotient of corresponding rows in column 'B' divided by B1. Here B\$1 is called **Absolute reference**.

That's about referencing "One Cell". Any ideas how we can refer a range or a group of cells in excel? Let's learn the same in next few slides.

In case, the reference used is incorrect then function will throw error and cell appears as "#REF"

Referencing a range

You can refer a complete range by defining it in the formulae bar inside a function. i.e. if we want to select the range from A1 to C3, we can simply type A1:C3.

PS: This range selection works only when written inside a function.

Now, if we want to find the summation of all the values in the cell range of A1:C3 in cell D3, then the formulae to be written will be “=sum(A1:C3)”

	A	B	C	D
1	23	3		
2		5	65	
3			24	=SUM(A1:C3)
4				

	A	B	C	D
1	23	3		
2		5	65	
3			24	120
4				

Result will be 120 in cell D3

Referencing a Cell

Coming back to Referencing a cell, there can be multiple ways of doing so, based on what we want to do?

In case, we want to have same logic applied to all the cells, i.e. if we want the cell in Column C to be a product of the values in respective

row values of column A & b, then the formulae will be $A_{<n>} * B_{<n>}$ where n is the row number. This is called Relative reference.

By default, reference is always a Relative one.

If we want each cell value to be multiplied by A1 (fixed cell) then the formulae will be $\$A\$1 * B_{<n>}$. This is called Absolute reference.

And if we mix both types of references, then it will be mixed one. Or, if we want either of the row or column to be fixed, and other one to be changed, then it will be mixed one.

Next slide, explain this better.

Relative Reference

- As explained in previous slides, relative reference is simply referring a cell from another.
- To enter a relative reference, type the cell reference as it appears in the worksheet. For example, enter B2 for referring cell B2, enter B3 for referring B3 and so on.
- Now, let's try an example of a trading strategy. In one of the strategies, we need to calculate the average closing price of a stock for last 3 days for a dataset of an year (i.e. we have the closing price data of a stock for an year).

- Image on right is a snapshot from the year which we need to calculate the average price for last three days.

Here, the formulae will be same, and can be Applied using relative reference.

Relative Reference

Formulae will be “=AVERAGE(B2,B3,B4)”

	A	B	C	D	E	F	G
1	SBI						
2	Date	Close	Returns	Average of last three days			
3	10-02-2017	276.35	0.001811				
4	11-02-2017	275.85	-0.00488				
5	12-02-2017	277.2	-0.00252	=AVERAGE(B3,B4,B5)			
6	13-02-2017	277.9	0.003063				

1	SBI		
2	Date	Close	Returns
3	10-02-2017	276.35	0.001811
4	11-02-2017	275.85	-0.00488
5	12-02-2017	277.2	-0.00252
6	13-02-2017	277.9	0.003063
7	14-02-2017	277.05	-0.0018
8	15-02-2017	277.55	0.015431
9	16-02-2017	273.3	0.009559
10	17-02-2017	270.7	0.038984
11	18-02-2017	260.35	-0.01373
12	19-02-2017	263.95	-0.00943
13	20-02-2017	266.45	0.027587

for
closing

If we copy and paste the formulae (Ctrl+C and Ctrl+V) in cell D4, then the formulae becomes “=AVERAGE(B4,B5,B6)”

	A	B	C	D	E
1	SBI				
2	Date	Close	Returns	Average of last three days	
3	10-02-2017	276.35	0.001811		
4	11-02-2017	275.85	-0.00488		
5	12-02-2017	277.2	-0.00252	276.4666667	
6	13-02-2017	277.9	0.003063	=AVERAGE(B4,B5,B6)	
7	14-02-2017	277.05	-0.0018	AVERAGE(number1, [number2], [number3])	

This is how Relative Reference works. Do try some more examples!

Absolute Reference

- Unlike, Relative reference, in Absolute Reference, the reference cell remains fixed; i.e. if we use Absolute reference in last example, for cell B3, then the

formulae will give average of B3, B4,B5 and then an average for B3,B5,B6 and so on.

- To enter an absolute reference, type \$ (a dollar sign) before both the row and column references. For example, enter \$B\$3 in last example. Result would be

	A	B	C	D
1	SBI			
2	Date	Close	Returns	Average of last three days
3	10-02-2017	276.35	0.001811	
4	11-02-2017	275.85	-0.00488	
5	12-02-2017	277.2	-0.00252	=AVERAGE(\$B\$3,B4,B5)

Absolute Reference

Now, when we move to next cell,

	A	B	C	D
1	SBI			
2	Date	Close	Returns	Average of last three days
3	10-02-2017	276.35	0.001811	
4	11-02-2017	275.85	-0.00488	
5	12-02-2017	277.2	-0.00252	276.4666667
6	13-02-2017	277.9	0.003063	=AVERAGE(\$B\$3,B5,B6)

Result will be

	A	B	C	D
1	SBI			
2	Date	Close	Returns	Average of last three days
3	10-02-2017	276.35	0.001811	
4	11-02-2017	275.85	-0.00488	
5	12-02-2017	277.2	-0.00252	276.4666667
6	13-02-2017	277.9	0.003063	277.15

277.15 with absolute reference

Mixed Reference

- There is one more kind of reference – “Mixed Reference”. As name says, it is a mix of relative and absolute i.e. one of the row or column is fixed while the other changes.
- To enter a mixed reference, type \$ before either the row or column reference. For example, enter \$B2 or B\$2
- If you want to keep column fixed and scroll through the row then use “\$” before column like \$B2 or if you want to keep row fixed and change the column, then use B\$2.
- Effectively, mixed reference is, you keep Row constant by using “\$” sign and change the column like B\$2, C\$2, D\$2, etc. or keep reference to Column constant by changing rows like \$B2, \$B3, \$B4, etc.

Referencing a Cell

Here is the summary, which will help you understand this better

- Relative Reference

- Used: Without '\$'

- Example: $C1 = 2 * B1$, $C2 = 2 * B2$ and so on. Column 'C' is double of column 'B'. OR

- Example: $C1 = A1 * B1$, $C2 = A2 * B2$ and so on. Column 'C' is product of columns 'A' and 'B'.

- Useful when you want to keep the same logic in calculation across the sheet

- Absolute Reference

- Used: To refer to a 'single' cell. '\$' is used for both row and column

- Example: $C1 = B1 * \$A\1 , $C2 = B2 * \$A\1 and so on. Column 'C' is A1 times of column 'B', whatever the value of A1 maybe.

- Useful when you want to use a single factor in calculations across the sheet.

- Mixed Reference

- Used: To refer to a column or a row. '\$' is used for either row or column

- Example: $C1 = B1 * \$A1$, $C2 = B2 * \$A2$ and so on. Column 'C' is product of columns 'A' and 'B'. OR
- Example: $C1 = B1 * A\$1$, $C2 = B2 * A\$1$ and so on. Column 'C' is A1 times of column 'B', whatever the value of A1 maybe.

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Referencing a Cell

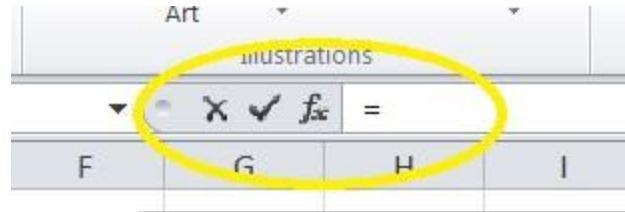
You can switch between the different references using a Short cut:

- Select the cell reference you want to change
- Press the key F4 to cycle the reference from relative to absolute to mixed and back to relative

Now, you understand how simple is to play with data in an Excel!

Formula

- **Formula** is an expression that returns a value
- Formula is written using **operators** that combine different values, returning a single value that is then displayed in the cell
- Alternatively, you can write a formula in Formula bar using “=”



- The most commonly used operators are **arithmetic operators**
- For example, to add the values in the range B1:B10, you could enter the following long formula:
$$=B1+B2+B3+B4+B5+B6+B7+B8+B9+B10$$

- Or, you could use the SUM function to accomplish the same thing: =SUM(B1:B10)
- Hence, Formula is a combination of one or more functions used in Excel.

Now, let's understand what a function is in Excel.

Functions

Excel can be used to execute mathematical functions over a varied range of data. Excel has a lot of inbuilt functions which can be used/combined to create formulae.

Typically there are about 10-12 types of functions, but most used types are:

- Mathematical functions
- Statistical functions
- Logical functions
- Lookup & Reference

Mathematical Functions

There are many mathematical functions available in Excel. Most commonly used among the Mathematical functions are:

- RAND
- LOG
- PRODUCT
- SUMIF
- COUNTIF

We will understand their meaning, what they do and how we can use them in upcoming slides.

RAND

- Returns a random number greater than equal to 0 and less than 1, evenly distributed.

`=RAND()`

- Can be used Syntax for RAND is

0.753326

- And the result is...
- We do not need any dataset or any reference cell value to use this function.

Log

- Returns the logarithmic value of the argument to the base you specify.
- Syntax is =LOG(a,b) where a is the number whose logarithmic value has to be calculated, and b is the base.
- The formulae for log value of 2 to the base 10 is

=LOG(2,10)



0.30103

- If you wish to refer any cell, then formulae will become =LOG(A2,B2) where value in cell A2 will be the number while B2 will be the base.

- One needs to make sure that both cell values are numbers.

Product

Returns the product of the arguments, can be numbers or cell names

	A	B	C	D	E	F	G
1	SBI						
2	Date	Close	Returns				
3	10-02-2017	276.35	0.001810939				
4	11-02-2017	275.85	-0.004882028				
5	12-02-2017	277.2	-0.002522069				
6	13-02-2017	277.9	0.003063341				
7	14-02-2017	277.05	0.001803102				
8	15-02-2017	277.55	0.015431004				
9	16-02-2017	273.3	0.009558896				
10	17-02-2017	270.7	0.038984317				
11	18-02-2017	260.35	-0.013732812				
12	19-02-2017	263.95	-0.009426917				
13	20-02-2017	266.45	0.027586644				
14	21-02-2017	259.2	0				
15	22-02-2017	259.2	0.016728653				
16	23-02-2017	254.9	0.002946667				

=PRODUCT(B3,B6,B9,B12)



5539994247

SUMIF

The SUM IF function is a mathematical function that returns the sum value of only those items that meet the specified condition.

The syntax of SUMIF function is :

`=SUMIF(range,criteria,[sum_range])`

- `'range'` refers to the cells that you want to be evaluated by the `'criteria'`.
- `'criteria'` refers to the condition that specifies which items are to be added. `'criteria'` can be a number, expression or a text string.
- `'sum_range'` is an optional argument, it specifies the cells to be added. If `'sum_range'` argument is omitted then SUMIF treats `'range'` as `'sum_range'`.

For example, C100 = (C1:C99, "buy", B1:B99)

C100 will be sum of values in column 'B' for which corresponding values in column 'C' are "buy". So for instance if 'B' consists of PnL, then C100 will have total PnL for all "buy" cases.

SUMIF Example

In below example, the values in column will be added only if they are greater than 0 else they will be treated as 0 during SumIf function execution

	A	B	C	D	E	F
1	SBI					
2	Date	Close	Returns			
3	10-02-2017	276.35	0.001810939			
4	11-02-2017	275.85	-0.004882028			
5	12-02-2017	277.2	-0.002522069			
6	13-02-2017	277.9	0.003063341			
7	14-02-2017	277.05	-0.001803102			
8	15-02-2017	277.55	0.015431004			
9	16-02-2017	273.8	0.009558896			
10	17-02-2017	270.7	0.038984317			
11	18-02-2017	260.35	-0.013732812			
12	19-02-2017	263.95	-0.009426917			
13	20-02-2017	266.45	0.027586644			
14	21-02-2017	259.2	0			
15	22-02-2017	259.2	0.016728653			
16	23-02-2017	254.1	0.002946667			

`=SUMIF(C3:C16, ">0")`

→

0.116110462

`=SUM(C3:C16)`

→

0.083743534

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PS: Sum is another Mathematical function which gives summation of all the arguments.

This function would be very useful when you have to add total profit (positive) and total loss (negative) for your strategy!

COUNTIF

The COUNTIF function counts the number of cells in a range, that meets a given criteria

SYNTAX:

=COUNTIF(range,criteria)

Here, 'range' refers to the range of cells that you want to be counted as nonblank cell if they meet the criterion.

'criteria' refers to the condition that specifies which items are to be counted.
'criteria' can be a number, expression or a text string.

COUNTIF Example

We count the number of times the price of the stock was greater than 275
. The answer implies the price was greater than 275 on 6 days.

	A	B	C	D	E	F	G
1	SBI						
2	Date	Close	Returns				
3	10-02-2017	276.35	0.001810939				
4	11-02-2017	275.85	-0.004882028				
5	12-02-2017	277.2	-0.002522069			=COUNTIF(B3:B16,">275")	6
6	13-02-2017	277.9	0.003063341			=COUNT(B3:B16)	14
7	14-02-2017	277.05	-0.001803102				
8	15-02-2017	277.55	0.015431004				
9	16-02-2017	273.3	0.009558896				
10	17-02-2017	270.7	0.038984317				
11	18-02-2017	260.35	-0.013732812				
12	19-02-2017	263.95	-0.009426917				
13	20-02-2017	266.45	0.027586644				
14	21-02-2017	259.2	0				
15	22-02-2017	259.2	0.016728653				
16	23-02-2017	254.9	0.002946667				

PS: Count is another Mathematical function which gives count of all the arguments.

Statistical Functions

One of the most commonly used Statistical function is “Average”. This function is used to calculate average value of a dataset.

- Syntax for this function is AVERAGE(number1, number2, etc.). You can write either cell name or the numerical value.
- Formulae for Average of 2 and 3 is =AVERAGE(2,3) , while formulae for average of cell B2 and B3 will be =AVERAGE(B2,B3)

Average

In the following image, we want to calculate the

average of column B.

	A	B	C	D	F	G
1	SBI					
2	Date	Close	Returns			
3	10-02-20	7 276.35	0.001810939			
4	11-02-20	7 275.85	0.004882028	=AVERAGE(B3:B16)	269.2821	
5	12-02-20	7 277.2	0.002522069			
6	13-02-20	7 277.9	0.003063341			
7	14-02-20	7 277.05	0.001803102			
8	15-02-20	7 277.55	0.015431004			
9	16-02-20	7 273.3	0.009558896			
10	17-02-20	7 270.7	0.038984317			
11	18-02-20	7 260.35	0.013732812			
12	19-02-20	7 263.95	0.009426917			
13	20-02-20	7 266.45	0.027586644			
14	21-02-20	7 259.2	0			
15	22-02-20	7 259.2	0.016728653			
16	23-02-20	7 254.9	0.002946667			

The formula becomes
=AVERAGE(B3,B4,B5,
B6,B7,B8,B9,B10,B11,
B12, B13, B14, B15,
B16)

Alternatively, we can
also write
=AVERAGE(B3:B16)

Logical Functions

- A logical function is a function that works with values that are either true or false
- Most Commonly used logical function is “IF” function
- The IF function is a logical function that returns one value if the condition is true and returns a different value if the condition is false
- Syntax is “=IF(logical_test, [value_if_true],[value_if_false])”

Logical Functions - Example

Let us proceed with an example for stock price data.

If the price on any day is higher than the mentioned price then we say “Sell”.

If the price on any day is less than the mentioned price then we say “Buy”.

Let's say the mentioned price is 275, so our strategy becomes, if Price is

	A	B	C
1	SBI		
2	Date	Close	Returns
3	10-02-2017	276.35	0.001810939
4	11-02-2017	275.85	-0.004882028
5	12-02-2017	277.2	-0.002522069
6	13-02-2017	277.9	0.003063341
7	14-02-2017	277.05	-0.001803102
8	15-02-2017	277.55	0.015431004
9	16-02-2017	273.3	0.009558896
10	17-02-2017	270.7	0.038984317
11	18-02-2017	260.35	-0.013732812
12	19-02-2017	263.95	-0.009426917
13	20-02-2017	266.45	0.027586644
14	21-02-2017	259.2	0
15	22-02-2017	259.2	0.016728653
16	23-02-2017	254.9	0.002946667

higher than 275, then one should “Sell” and if Price is less than 275, then “Buy”.

So, how to define this logic in excel?

We will see the same in next slide!

Logical Functions – Syntax

=IF(B3>275,“Sell”,“Buy”)

The above command says that if the value in cell B3 is greater than 275 then as per our condition we should Sell else we should Buy.

Column B contains the closing price for each day

	A	B	C	D	E	F	G
1	SBI						
2	Date	Close	Returns				
3	10-02-2017	276.35	0.001810939				
4	11-02-2017	275.85	-0.004882028		=IF(B3>275,"Sell","Buy")		
5	12-02-2017	277.2	-0.002522069				
6	13-02-2017	277.9	0.003063341				
7	14-02-2017	277.05	-0.001803102				
8	15-02-2017	277.55	0.015431004				
9	16-02-2017	273.3	0.009558896				
10	17-02-2017	270.7	0.038984317				
11	18-02-2017	260.35	-0.013732812				
12	19-02-2017	263.95	-0.009426917				
13	20-02-2017	266.45	0.027586644				
14	21-02-2017	259.2	0				
15	22-02-2017	259.2	0.016728653				
16	23-02-2017	254.9	0.002946667				

Look up Function

- A lookup table is a table that organizes data you want to retrieve into different categories
- The categories for the lookup table, called compare values, are located in the table's first column or row

- To retrieve a particular value from the table, a lookup value (the value you are trying to find) needs to match the compare values
- The formula for lookup function is
`=VLOOKUP(lookup_value,table_array,col_index_num,[range_lookup])` where lookup_value is the value to search in firstcolumn of the excel
table_array: column of excel sorted in ascending order
col_index_num: The column number from which the matching value must be returned. Here, firstcolumn is 1
range_lookup: logical value. To find the closest match, Keep as “True”

Look up Function

Here, in the example, we have two dataset – Closing price and returns for two stocks – ABC and XYZ. We want to know what is the return for “XYZ” stock on 10 Feb 2017. Let’s use VLOOKUP function to find the same.

The formulae becomes =VLOOKUP(A3,E3:G16,2,0); where A3 is the common value between both datasets; while E3:G16 is the range ; and 2,0 imply that we need the formula to return the value of cell which is on 2nd positions from the date value and in same row.

	A	B	C	D	E	F	G	H	I	J	K
1	ABC				XYZ						
	Date	Close	Returns		Date	Close	Returns				
3	10-02-2017	276.35	0.001810939		23-02-2017	265.7	0.020415		=VLOOKUP(A3,E3:G16,2,0)		
4	11-02-2017	275.85	-0.004882028		18-02-2017	269.35	0.027309		VLOOKUP(lookup_value,tabl		
5	12-02-2017	277.2	-0.002522069		17-02-2017	270.7	-0.00317				
6	13-02-2017	277.9	0.003063341		22-02-2017	273.5	-0.00995				
7	14-02-2017	277.05	-0.001803102		19-02-2017	273.95	0.004987				
8	15-02-2017	277.55	0.015431004		12-02-2017	275.2	-0.00435				
9	16-02-2017	273.3	0.009558896		20-02-2017	276.45	-0.01708				
10	17-02-2017	270.7	0.038984317		15-02-2017	277.42	-0.00026				
11	18-02-2017	260.35	-0.013732812		10-02-2017	277.45	0.000164				
12	19-02-2017	263.95	-0.009426917		13-02-2017	277.9	0.016357				
13	20-02-2017	266.45	0.027586644		14-02-2017	278.05	0.000164				
14	21-02-2017	259.2	0		11-02-2017	278.24	0.003475				
15	22-02-2017	259.2	0.016728653		16-02-2017	278.3	0.002266				
16	23-02-2017	254.9	0.002946667		21-02-2017	279.2	-0.00913				
17											

The result is

277.45

Vlookup is helpful when dataset is huge.

OFFSET Function

The Offset function in Excel returns a cell or range of cells that is at a specified number of rows and columns from a cell or range of cells.

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Syntax is:

=OFFSET(cell,no_rows,no_columns,height,width)

where

Inputting height and width are optional

No_rows is number of rows from cell (positive means below and negative refers to above)

No_columns is number of columns from the cell (positive means towards right and negative means on the left).

For example, “=OFFSET(B2,6,4)” will refer to cell which is 6 rows below Row 2 (i.e. Row 8) and 4 columns on right of Column B (i.e. Column F) which is cell F8.

Similarly, OFFSET(G10,-2,-5) will refer to cell which is 2 rows above row 10 (i.e. Row 8) and 5 columns on left of Column G (i.e. Column B) which is cell B8.

OFFSET Function

In the following example, we see the result of function is the price of the stock three days after 10/02/2017 (which is 277.9 on 13/02/2017) using the offset function.

	A	B	C	D	E	F	G
1	ABC						
2	Date	Close	Return				
3	10-10-2017	276.35	0.001810939				
4	11-10-2017	275.85	-0.004882028				
5	12-10-2017	277.2	-0.002522069				
6	13-10-2017	277.9	0.003063341				
7	14-10-2017	277.05	-0.001803102				
8	15-10-2017	277.55	0.015431004				
9	16-10-2017	273.3	0.009558896				
10	17-10-2017	270.7	0.038984317				
11	18-10-2017	260.35	-0.013732812				
12	19-10-2017	263.95	-0.009426917				
13	20-10-2017	266.45	0.027586644				
14	21-10-2017	259.2	0				
15	22-10-2017	259.2	0.016728653				
16	23-10-2017	254.9	0.002946667				
17							

=OFFSET(A3,3,1,1,1)

➔

277.9

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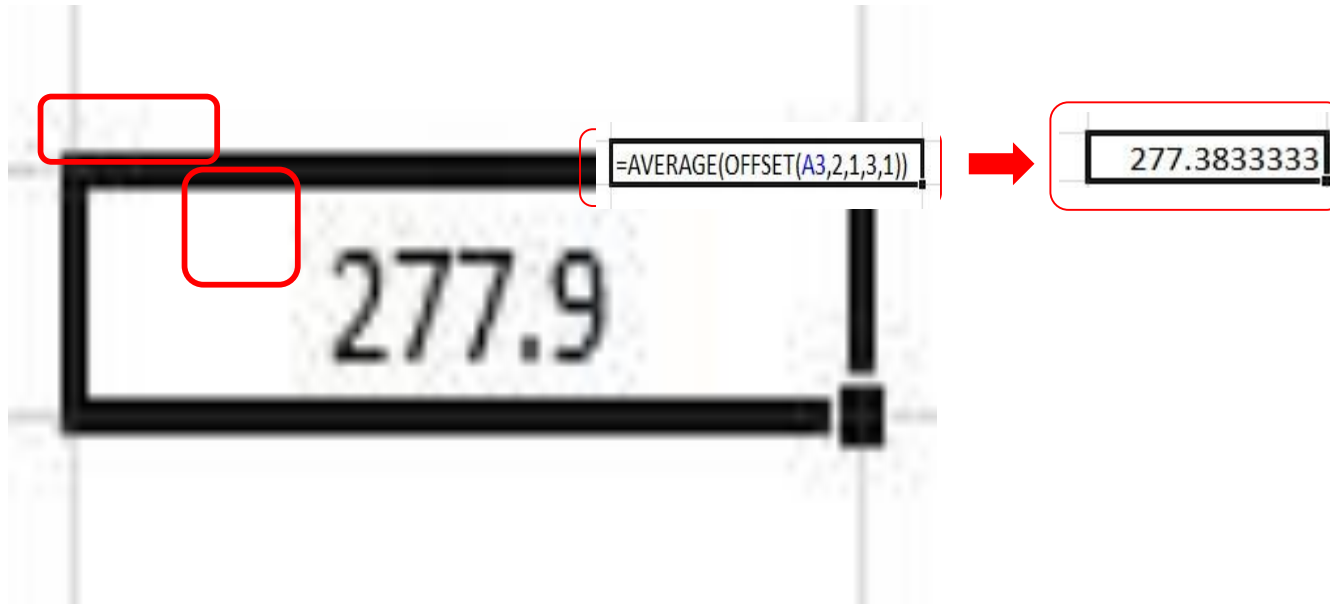
Here, height and width are set to 1. Do you know why?

OFFSET Function

Height and Width determine the result set. In case, we want a result with only one cell, then both height and width are set to 1

If we want result to be the average of the three cells, then we set height as 3

and width as 1.



Similarly, we can include the next column by increasing width. So now, you know what would be the result of `=PRODUCT(OFFSET(A3,2,1,2))`?

INDEX

If the dataset is huge, then you may encounter performance issues with OFFSET. In such a scenario, INDEX is the function you can use.

The INDEX function returns either the cell value or the reference to a cell value from a table or range. The syntax is:

`=INDEX(array,row_num,column_num)`

Where

Array is the complete range for which the reference is given

Row_num is number of row above (when negative) or below (when positive) the reference

Column_num is the number of columns on the left (when negative) or the right (when positive) of the reference

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Let's take an example to understand this better.

INDEX

Here, in the given example, we want to fetch the value of cell C4 by using INDEX function.

To do so, the function will be `=INDEX(A2:C16,3,3)` where A2:C16 define the range of data, while 3 defines 3rd row from the reference and the next 3

	A	B	C	D	E	F
1	ABC					
2	Date	Close	Returns			
3	10-02-2017	276.35	0.001810939			
4	11-02-2017	275.85	-0.004882028			
5	12-02-2017	277.2	-0.002522069			
6	13-02-2017	277.9	0.003063341			
7	14-02-2017	277.05	-0.001803102			
8	15-02-2017	277.55	0.015431004			
9	16-02-2017	273.3	0.009558896			
10	17-02-2017	270.7	0.038984317			
11	18-02-2017	260.35	-0.013732812			
12	19-02-2017	263.95	-0.009426917			
13	20-02-2017	266.45	0.027586644			
14	21-02-2017	259.2	0			
15	22-02-2017	259.2	0.016728653			
16	23-02-2017	254.9	0.002946667			

`=INDEX(A2:C16,3,3)`

INDEX(array, row, column)
INDEX(reference, row, column)

The result is as shown in below image :

Which is the value in column C4

-0.004882028

range of data, while 3 defines 3rd row from the reference and the next 3

Until now, we have learnt,

- Different kinds of functions used in Excel, how to use them
- Different charts available in Excel
- Relevant charts for Quantitative and Algorithmic Trading
- How to work with Data in excel

This is all for PRIMER- Excel.

Next, you will learn how to use these basic concepts in creating a Trading Strategy in Excel in the EPAT lectures!

Till then, keep learning and practicing.